

E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models

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Abstract—Large language models (LLMs) have shown significant potential in guiding embodied agents to execute language instructions across a range of tasks, including robotic manipulation and navigation. However, existing methods are primarily designed for static environments and do not leverage the agent’s own experiences to refine its initial plans. Given that real-world environments are inherently stochastic, initial plans based solely on LLMs’ general knowledge may fail to achieve their objectives, unlike in static scenarios. To address this limitation, this study introduces the Experience-and-Emotion Map (E2Map), which integrates not only LLM knowledge but also the agent’s real-world experiences, drawing inspiration from human emotional responses. The proposed methodology enables one-shot behavior adjustments by updating the E2Map based on the agent’s experiences. Our evaluation in stochastic navigation environments, including both simulations and real-world scenarios, demonstrates that the proposed method significantly enhances performance in stochastic environments compared to existing LLM-based approaches. The code and supplementary materials are available at <https://e2map.github.io/>.

I. INTRODUCTION

Large language models (LLMs), pre-trained on Internet-scale data, have emerged as a promising method to encapsulate the world’s knowledge distilled in language. These LLMs have demonstrated a variety of capabilities, including interpreting and responding to natural language instructions, performing logical reasoning, and generating code. Leveraging these capabilities, many studies have explored applying the generalizable knowledge of LLMs to embodied agents, enabling them to interact physically in the real world.

To effectively utilize LLMs’ knowledge in enabling embodied agents, it is crucial to ground this knowledge in the real-world environments where the agents operate. Previous studies have proposed methods to decompose language instructions into sequential subtasks, which are then executed using predefined motion primitives [1]–[3]. Other research has explored the use of LLMs’ code generation capabilities to translate language instructions into executable code for robots, which is then used in conjunction with various APIs [4]–[6]. Additionally, some studies have focused on grounding information from LLMs and vision-language models (VLMs) in the spatial contexts where robots operate [7]–[9].

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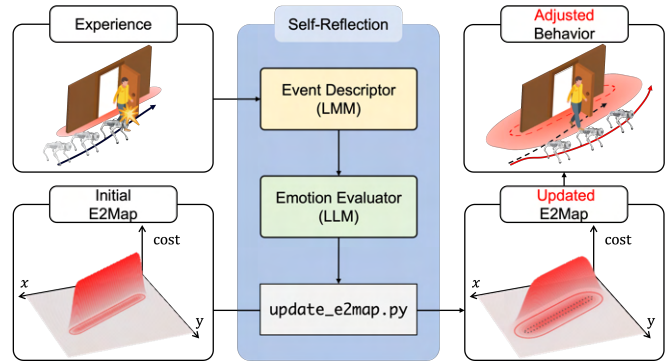


Fig. 1. E2Map is a spatial map that captures the agent’s emotional responses to its experiences. Our method enables one-shot behavior adjustments in stochastic environments by updating the E2Map through the diverse capabilities of LLMs and LMMs. Since VLM can refer to both the Vision-Language Model and Visual Language Map, we will preferably use LMM for the former and VLM for the latter.

For instance, [7] proposed a method for planning by representing language instructions as spatial costs and affordances using LLMs’ code-writing capabilities. Furthermore, [8] and [9] introduced methodologies for robot navigation that use maps that allow spatial information queries by language.

While these methodologies have successfully demonstrated the use of LLMs with embodied agents, they primarily address static environments and fail to incorporate the agent’s own experiences. Given that the real world is inherently stochastic and subject to various uncertainties, initial plans based solely on the general knowledge of LLMs may fall short of achieving their objectives. For example, a robot navigating indoors might unexpectedly collide with a person stepping out from a suddenly opened door, potentially causing it to fail. In such cases, the robot may fail to complete its mission using its initial plan. To increase the success rate in similar future scenarios, it is essential to incorporate these experiences and adjust the robot’s behavior to maintain a safe distance when encountering doors. In summary, to achieve objectives in stochastic environments, the robot must refine its plan based on experience.

Humans experience emotions in response to stimuli, which influence specific behaviors toward certain objects or situations. According to [10] and [11], emotions play a crucial role in maintaining homeostasis by guiding humans to avoid previously encountered dangers, thus supporting survival. Furthermore, recent research [12] demonstrates that humans integrate emotional experiences into spatial representations

for navigation. By using emotions as spatial boundaries to maintain homeostasis, humans can prefer or avoid specific locations based on their experiences during navigation.

Inspired by human emotional mechanisms, this study proposes a language-based robot system that utilizes the capabilities of LLMs and large multi-modal models (LMMs) to enable an embodied agent to autonomously adjust its behavior in a one-shot manner. The core component is the Experience-and-Emotion Map (E2Map), which serves as a spatial map that grounds both the general knowledge of LLMs and the agent's emotional responses to experiences. When the agent interacts with a particular object or space, it quantifies its emotional response and integrates this information into the E2Map, as shown in Fig. 1. The agent's planning and control module then uses the E2Map as a cost function to guide its actions, allowing for behavior refinement through E2Map updates. The contributions are summarized as follows:

- Inspired by human emotional mechanisms, we propose E2Map, a spatial representation that integrates both the general knowledge of LLMs and the agent's emotional responses to its experiences to facilitate planning.
- We propose a robot system that leverages the E2Map and various LLM capabilities—including code generation, event description, and emotion evaluation—to enable the agent to autonomously adjust its behavior in a one-shot manner based on its experiences.
- We verified that the proposed methodology can autonomously adjust behavior by incorporating its experiences in navigation environments with changing conditions and dynamic objects, demonstrating superior performance over existing LLM-based approaches.

II. RELATED WORKS

A. LLM-based Robotics

Research on using language instructions to control robots has advanced significantly with the development of LLMs. Numerous studies [1]–[7], [13]–[17] have aimed to leverage the general knowledge embedded in LLMs and the benefits of few-shot prompting, which eliminates the need for additional training. [1]–[3] proposed a method that uses LLMs to decompose a given language instruction into a sequence of subtasks and employs predefined motion primitives to execute each subtask. [4]–[6] utilized LLMs' capability of code generation to transform language instructions into executable code for robots, using various APIs to carry out tasks. [7] proposed utilizing LLMs to construct affordances and constraints within a 3D voxel map and employing model predictive control (MPC) to compute actions, eliminating the need for predefined motion primitives.

While these approaches have shown notable success in language-based robotics, they primarily focus on static environments and overlook the integration of the robot's experiences. [17] demonstrated that it is possible to refine a robot's behavior through additional human language instructions, but this refinement does not stem from the robot's own experiences. In contrast, the proposed methodology addresses

TABLE I
COMPARISON OF EXISTING AND PROPOSED METHODS.

Methods	LLM-Based	Spatial Grounding	Behavior Adjustment	Self-Reflection
[1]–[6], [18]	✓			
[7]–[9]	✓	✓		
[17]	✓		✓	
Ours	✓	✓	✓	✓

behavior refinement in stochastic environments by grounding the robot's emotional responses to its own experiences.

B. Visual-Language Navigation

To demonstrate the ability to achieve one-shot behavior adjustment in stochastic environments, we evaluated the proposed methodology in a mobile robot indoor navigation scenario. Navigation environments involve dynamic objects such as people, which can cause unexpected events, e.g., a person suddenly stepping out of the door.

Visual-language navigation (VLN), which uses a robot's visual input to execute given language instructions, has been extensively researched to date. Early research used methods to map language instructions and visual input to discrete [19], [20], or continuous actions [21], [22]. However, these methods are data-intensive, which raises concerns about the cost of data collection. To address these problems, research has emerged that uses foundation models [8], [9], [18], [23]. [23] proposed a topological graph-based navigation approach using three different foundation models. [18] demonstrated object navigation using open-vocabulary object detector based on CLIP [24] and exploration techniques. [8] proposed creating a prebuilt spatial map that stores visual-language features of objects, while [9] suggested generating maps by storing pixel-level visual-language features of spaces. By grounding language instructions into these maps, both approaches demonstrated their effectiveness in enabling spatial open-vocabulary navigation tasks. However, these approaches are limited to static environments and do not account for the refinement of navigation plans based on the agent's real-time experiences. Table I presents a comparison between the proposed method and existing methods.

C. Learning from Experience

Methods for optimizing robot behavior based on past experiences have been widely explored in reinforcement learning [25]. Reinforcement learning quantifies rewards from experiences and trains models to maximize these rewards. Similarly, in imitation learning, several studies have proposed acquiring additional expert data when the agent experiences a situation it cannot solve on its own during operation [26], [27]. However, these approaches require extensive interactions with the environment or additional data acquisition for learning, making immediate behavior corrections impractical. In contrast, the proposed method leverages LLM prompting, avoiding parameter updates, and grounds the agent's experiences in E2Map for immediate behavior corrections in a one-shot manner.

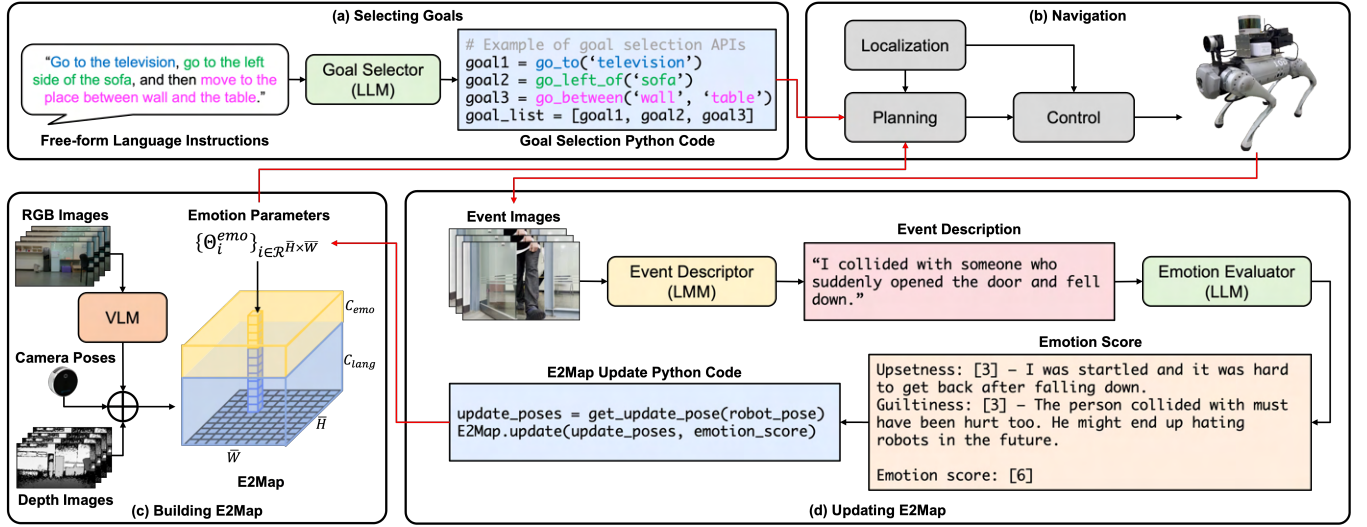


Fig. 2. System Overview: (a) E2Map is created by embedding visual-language features and emotion parameters into corresponding grid cells. (b) When a user provides a language instruction, the *goal selector* generates code through goal selection APIs to decide goals to reach. (c) The planning algorithm then uses emotion as a cost to generate the optimal path to the goal. (d) If the agent encounters unexpected events during navigation, the E2Map is updated through the sequential operation of the *event descriptor* and *emotion evaluator*. Following the update, the planning algorithm replans the path to adjust the agent's behavior in a one-shot manner.

III. METHOD

A. Building and Initializing E2Map

E2Map is a spatial grid map that captures both the visual-language features of the environment and the agent's emotional responses to its experiences. In this paper, we define emotion as the spatial extent reflecting how the agent perceives the space to maintain homeostasis, inspired by [10]. We ground emotion within the spatial map by modeling it as a weighted summation of Gaussian distribution.

E2Map is mathematically defined as $\mathcal{M} \in \mathcal{R}^{\bar{H} \times \bar{W} \times (C_{lang} + C_{emo})}$, where $\bar{H}$ and $\bar{W}$ represent the size of the top-down grid map, C_{lang} denotes the dimension of the visual-language features for each grid cell, and C_{emo} indicates the number of parameters for emotion. Each grid cell represents s meters of actual space, so the size of the space represented by E2Map is $s\bar{H} \times s\bar{W}$ meters. The proposed methodology first builds $\mathcal{M}_{lang} \in \mathcal{R}^{\bar{H} \times \bar{W} \times C_{lang}}$ by embedding visual-language features of RGB images of the environment into a corresponding grid cell. Similar to [9], we used LSeg [28] as the pre-trained VLM to encode RGB images and determine the position of each pixel in the grid map using depth images and camera poses.

To embed the emotion for each grid cell, we first define a multivariate Gaussian distribution $\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_{\mathbf{p}_i}, \boldsymbol{\Sigma}_{\mathbf{p}_i})$ for each occupied grid cell $\mathbf{p}_i = [x, y]^T$, where $\mathbf{x} \in \mathcal{R}^{\bar{H} \times \bar{W}}$, $\boldsymbol{\mu}_{\mathbf{p}_i} = [x, y]^T$ and $\boldsymbol{\Sigma}_{\mathbf{p}_i} = [\sigma_x^2, 0; 0, \sigma_y^2]$. The covariance matrix is initialized as an identity matrix scaled by a coefficient. In the remainder of the paper, we use the terms $\mathcal{N}(\cdot | \boldsymbol{\mu}_{\mathbf{p}_i}, \boldsymbol{\Sigma}_{\mathbf{p}_i})$ and $\mathcal{N}_{\mathbf{p}_i}(\cdot)$ interchangeably to denote the same distribution. Finally, the emotion is calculated as the weighted summation of multivariate Gaussian distributions, as follows:

$$E(\mathbf{x}) = \sum_{\mathbf{p}_i \in \mathcal{O}} w_{\mathbf{p}_i} \mathcal{N}_{\mathbf{p}_i}(\mathbf{x}), \quad (1)$$

where $\mathcal{O}$ is the set of occupied grid cells, and $w_{\mathbf{p}_i}$ is the

weight parameter. Note that, to reduce computation time for calculating the emotion at the grid cell, we only consider Gaussian distributions whose value at the grid cell exceeds a specified threshold. The weight parameter $w_{\mathbf{p}_i}$ is initialized based on the number of these valid Gaussian distributions:

$$w_{\mathbf{p}_i} = \frac{1}{N_{\mathbf{p}_i}^{val}}, \quad \forall \mathbf{p}_i \in \mathcal{O}. \quad (2)$$

We initialize the weight as the reciprocal of the number of valid Gaussian distributions $N_{\mathbf{p}_i}^{val}$ to ensure that the emotion at the grid cell is not disproportionately influenced by the number of these distributions. The covariance matrix and the weight parameter constitute the emotion parameters $\theta_i^{emo} = \{\boldsymbol{\Sigma}_{\mathbf{p}_i}, w_{\mathbf{p}_i}\}$ for each occupied grid cell. These parameters are stored in $\mathcal{M}$ along with the visual-language feature and updated based on the agent's experiences.

B. Reflecting Emotion and Updating E2Map

The proposed methodology leverages the capabilities of LLM and LMM to update the E2Map based on the agent's navigation experiences. First, when the agent encounters an event, the *event descriptor*, an LMM, generates a language description that explains the image sequence of the situation. To narrow the scope of the problem, we assume that the agent's events are indicated through a simulator or sensor information, rather than relying on a separate event detection algorithm. Then, an *emotion evaluator*, an LLM, assesses the agent's emotional response to the experience. The *emotion evaluator* takes the language description of the situation as input and evaluates the emotion as a score based on two criteria. Finally, E2Map is updated considering the emotion score and the grid cells' location to update. The proposed methodology uses GPT-4o [29] for the *event descriptor* and Llama3 [30] for the *emotion evaluator*. A detailed explanation of each process is provided below.

You are a robot dog navigating in an indoor environment where humans are present. Based on a description of the event you experience, evaluate the emotions you feel. The desired traits for evaluation are: 1) Upsetness – the impact you feel in the given situation, and 2) Guiltiness – the impact your actions have on the environment.

Here are a few examples.

#1
 Event description: I accidentally bump into a coffee table, causing a cup to spill onto the floor.
 Scores:
 Upsetness: [2] – The situation disrupts my navigation and creates an unexpected obstacle. Although the impact is not severe, it requires me to reassess my path and avoid the spilled liquid to prevent further issues
 Guiltiness: [3] – My action directly caused a mess in the environment, potentially leading to inconvenience for the humans present. This could result in additional cleaning or concern from them.

Emotion score: [5]
 ...

Fig. 3. An example of a few-shot prompt for the *emotion evaluator*: The sentences highlighted in gray provide instructions to inform the LLM about the task. The sentence highlighted in green serves as an event description, while the sentences highlighted in blue demonstrate how to evaluate emotion. The full prompts for the *event descriptor* and *emotion evaluator* are provided in the supplementary material.

1) *Event descriptor*: The *event descriptor* generates a language description that explains the sequence of images of the agent’s experience. To enable the LMM to describe the situation, we input three images to the model: the image captured before the event $I_{t_{evt}-h}$, the image at the time of event $I_{t_{evt}}$, and the image after the event $I_{t_{evt}+h}$, where t_{evt} is the timestep when the event occurred and h is the hyperparameter. The LMM is then prompted with these images to generate a description of the event. To enable the LMM to provide a detailed description of the situation, we employ step-by-step reasoning, known for its strong performance in zero-shot prompting [31]. Specifically, instead of merely requesting a single description for the three images, we prompt the LMM by first indicating that the images represent sequential scenes of an event, requesting it to describe each scene individually and then combine them into a comprehensive explanation of the entire situation.

$$l_{evt} = \mathcal{F}_{ed}(I_{t_{evt}-h}, I_{t_{evt}}, I_{t_{evt}+h}, p_{ed}), \quad (3)$$

where l_{evt} is the description of the event, $\mathcal{F}_{ed}$ is the *event descriptor*, and p_{ed} is the prompt of the *event descriptor*.

2) *Emotion evaluator*: The *emotion evaluator* assesses the agent’s emotional response based on the language description of the situation generated by the *event descriptor*. To enable the LLM to evaluate the emotions related to the situation, we utilize few-shot prompting. An example of manually created few-shot prompts is shown in Fig. 3. We defined two criteria for evaluating emotions: *upsetness*, which represents the impact the agent feels in the given situation, and *guiltiness*, which represents the impact on the environment caused by the agent’s actions. Our criteria are inspired by human emotions that drive avoidance and reparative behavior [32]–[34]. We employed few-shot prompting to instruct the LLM to provide scores out of three for each criterion and to explain the reasons behind these scores. The final emotion score is computed by adding the two scores.

$$s_{emo} = \mathcal{F}_{ee}(l_{evt}, p_{ee}), \quad (4)$$

where s_{emo} is the emotion score, $\mathcal{F}_{ee}$ is the *emotion evaluator*, and p_{ee} is the few-shot prompt of *emotion evaluator*. Note that, although this study addressed negative emotions such as *upsetness* and *guiltiness*, demonstrations of the *emotion evaluator* for assessing positive emotions are provided in the supplementary material.

To update the E2Map, we first identify the locations of the n grid cells, denoted as $\mathbf{p}_{update}$, associated with the object or space involved in the event, taking into account the agent’s pose at the time of the event. Next, we calculate the unit direction vector $\vec{v} = (v_x, v_y)$ representing the agent’s orientation at the time of the event. This vector is used to adjust the emotion parameters of each relevant grid cell based on the emotion score. We then update the standard deviation of each diagonal element in the covariance matrix by incorporating both the emotion score and the corresponding component of the direction vector. Our update rule is inspired by the Weber–Fechner law [35], which states that the intensity of a sensation increases as the logarithm of the stimulus. The standard deviation is updated as follows:

$$\sigma_k^{new} = \sigma_k v_k \log\left(\frac{s_{emo}}{T}\right), \text{ where } k \in \{x, y\}, \quad (5)$$

$(\cdot)^{new}$ refers to the updated value or function, and T is the temperature parameter.

By updating the covariance matrix, the multivariate Gaussian distribution expands, which means the emotional response to the event affects a broader spatial area. However, since a multivariate Gaussian distribution is a probability density function that integrates to one over all regions, increasing the covariance reduces the value of the distribution at the grid cell’s location. This results in a smaller emotion value at the grid cell according to Eq. (1). To ensure that the emotion at the grid cell is maintained even after the covariance update, we propose a simple method for updating the weight parameter $w_{\mathbf{p}_i}$ as follows:

$$w_{\mathbf{p}_i}^{new} = w_{\mathbf{p}_i} \frac{\mathcal{N}_{\mathbf{p}_i}(\mathbf{p}_i)}{\mathcal{N}_{\mathbf{p}_i}^{new}(\mathbf{p}_i)}. \quad (6)$$

By updating the emotion parameters according to Eq. (5) and Eq. (6), we can expand the spatial extent of the emotion’s influence by recalculating Eq. (1), while preserving the magnitude of the emotion within the given grid cell.

C. Navigating with E2Map

In this section, we explain how the proposed system works using E2Map. The overall algorithm is presented in Algorithm 1. When a language instruction l_{inst} is received from the user, the *goal selector* converts this instruction into code that utilizes the goal selection APIs to determine the goal locations (line 1), as shown in Fig. 2-(b). The goal selection API returns the location of the grid cell that the robot needs to visit, considering the object mentioned in the language instruction and the spatial information related to that object. A detailed explanation of the *goal selector* is provided in the supplementary material.

Once the goal is determined, the agent can use any cost-based off-the-shelf navigation system to move towards the

Algorithm 1 Navigation with E2Map in the real world.

```

1: goal_list = GoalSelector( $l_{inst}$ )
2:  $p_{agent}$  = Localization( $P_0$ )
3: for goal in goal_list do
4:    $\tau$  = Planning( $p_{agent}$ , goal,  $\mathcal{M}$ )
5:   while not arrived(goal) do
6:      $a_t$  = MPPI( $p_{agent}$ ,  $\tau$ )
7:     agent.run( $a_t$ )
8:      $p_{agent}$  = Localization( $P_t$ )
9:     if event occurs then
10:       $l_{evt} = \mathcal{F}_{ed}(I_{t_{evt}-h}, I_{t_{evt}}, I_{t_{evt}+h}, p_{ed})$ 
11:       $s_{emo} = \mathcal{F}_{ee}(l_{evt}, p_{ee})$ 
12:       $p_{update}$  = get_update_pose( $p_{agent}$ )
13:       $\mathcal{M}$ .update( $s_{emo}$ ,  $p_{update}$ )
14:       $\tau$  = Planning( $p_{agent}$ , goal,  $\mathcal{M}$ )
15:    end if
16:     $t = t + 1$ 
17:  end while
18: end for

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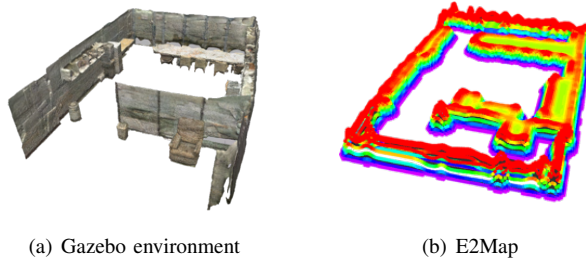


Fig. 4. We evaluated our method in the ROS Gazebo simulator. (a) We scanned the real-world setup using a 3D scanner and transferred the 3D model to the Gazebo simulator. (b) The initial E2Map for the environment.

goal. We use the D* algorithm [36] for path planning, MPPI [37] for control, and 3D LiDAR-based localization [38] for pose estimation. The agent's initial pose is estimated using LiDAR point cloud P_0 (line 2), and D* algorithm generates a minimum-cost path τ to the destination, using the emotion computed by Eq. (1) as a cost (line 4). The MPPI algorithm then calculates the control values a_t to follow this path, based on the agent's current position p_{agent} (line 6). During the agent's movement, a localization algorithm estimates the agent's position in real time (line 8).

If the agent encounters an event during navigation, the E2Map is updated (line 10–13), as described in Section III-B. When the navigation cost (emotion) changes, the D* algorithm recalculates a new minimum-cost path (line 14). This allows the agent to adjust its behavior in a one-shot manner based on a single experience.

Our method with E2Map appears similar to a potential field [39], where negative emotions act like repulsive forces and the goal functions as an attractive force. However, to circumvent the local minima problem inherent in potential fields, we employed a separate planner to determine the path toward the goal, instead of using the goal as the attractive force within the potential field.

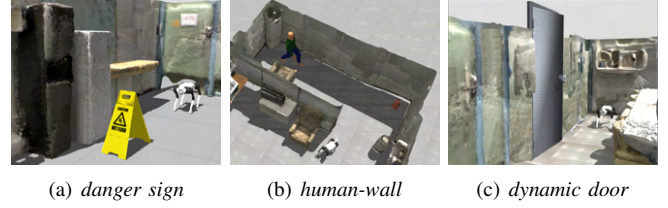
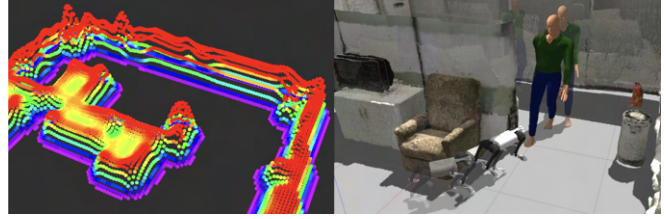
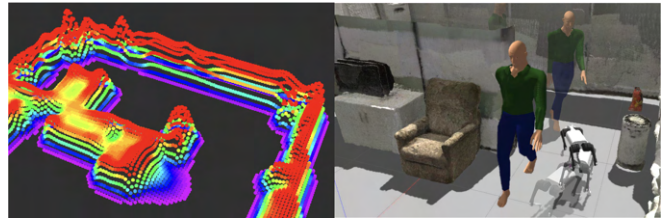


Fig. 5. Three experimental scenarios.



(a) Before the event



(b) After the event

Fig. 6. Update of the E2Map and behavior adjustment after an event: The spatial extent of the wall's emotional influence expands following incidents, such as a collision with a person stepping out from behind the wall. After the E2Map is updated, the robot adjusts its behavior to maintain a safe distance from the wall, preventing future collisions with humans.

IV. EXPERIMENTS

A. Experimental Setup

We conducted experiments in both simulated and real-world environments. Using the ROS Gazebo simulator [40], we created a simulated environment that mirrored the real-world setup used for evaluation, as shown in Fig. 4. In the simulated environment, we designed three scenarios to assess our method (Fig. 5). First, after building the initial E2Map, we introduced a new static obstacle, such as a danger sign (*danger sign*), to evaluate the method's ability to adapt to environmental changes. Second, we positioned a human figure behind a wall and had a human step out unexpectedly (*human-wall*). Third, we added a door that opened unexpectedly as the robot approached (*dynamic door*). The *human-wall* and *dynamic door* scenarios were designed to test whether our method could adjust behavior based on experiences with dynamic events.

For quantitative analysis, we compared our method against baselines with state-of-the-art performance in open-vocabulary object navigation [9], [23]. Our method and the baselines received identical language instructions and navigated accordingly, with success rates calculated for performance evaluation. Details about the baseline methods are provided in the supplementary material. After evaluation in the simulated environment, we demonstrated the scalability and applicability of our method in real-world scenarios.

TABLE II
QUANTITATIVE RESULTS IN SIMULATED ENVIRONMENT.

Methods	<i>static</i>	<i>danger sign</i>	<i>human-wall</i>	<i>dynamic door</i>
LM-Nav [23]	5/10	1/10	0/10	1/10
VLMs [9]	10/10	0/10	0/10	0/10
E2Map (ours)	10/10	9/10	9/10	9/10

B. Experiments in Simulated Environment

In the simulated environment, we provided ten different language instructions for both our method and the baselines for each of the three scenarios and calculated the success rate. For each episode, the robot started from the same position, and the language instructions referred to a maximum of four objects. Full language instructions are provided in the supplementary material. Success was defined as the robot reaching all goals in order without collisions, with a goal considered reached if the robot was within a specified distance from it. The episode was reset if a collision occurred or the agent reached the goal. Table II showed the results.

We first evaluated our method and the baselines in a *static* environment, without environmental changes or dynamic events, to assess their baseline performance using language instructions from the *danger sign* scenario. As shown in the table, both our method and VLMs succeeded in all ten attempts, while LM-Nav succeeded only five times. The challenges LM-Nav faces in multi-goal navigation were previously noted in [9]. In scenarios involving environmental changes and dynamic events, our method outperforms baselines in all three scenarios, as shown in the table. We found that the baselines repeatedly collided with the danger sign, human figure, and dynamic door, despite having previously encountered similar situations. In some cases, LM-Nav succeeded in the *danger sign* and *dynamic door* scenarios because its topological graph-based goal selection algorithm occasionally chose goal positions where the robot avoided encountering the danger sign and the door. In contrast, in the *danger sign* scenario, while our method initially collided with the danger sign, it avoided it in subsequent encounters by updating the E2Map. Similarly, in the *human-wall* and *dynamic door* scenarios, after experiencing collisions with the human figure and dynamic door, our method adapted its behavior to maintain a safe distance from both the wall and the door when passing by them. The qualitative result of our method in *human-wall* scenario is provided in Fig. 6. The results demonstrated that our method can adjust the agent's behavior in a one-shot manner, leading to a higher success rate in scenarios with environmental changes and dynamic events. Examples of qualitative results from the *event descriptor* and *emotion evaluator* for each scenario are provided in the supplementary material.

C. Experiments in Real World

To evaluate the scalability and applicability of our method in real-world settings, we first set up a real-world environment by placing objects such as a sofa, chair, table, refrigerator, and microwave in the conference room at Seoul National University, as shown in Fig. 7. We used the same



Fig. 7. Real-world setup for the experiments.

TABLE III
QUANTITATIVE RESULTS IN REAL-WORLD ENVIRONMENT.

Methods	<i>static</i>	<i>danger sign</i>	<i>human-wall</i>
E2Map (ours)	10/10	9/10	9/10

language instructions as in the simulation and incorporated real humans and danger signs to replicate the *danger sign* and *human-wall* scenarios from the simulation. We excluded the *dynamic door* scenario to prevent possible damage to the robot and the conference room door from collisions. As shown in Table III, our method demonstrated performance similar to that in the simulated environment, indicating its potential for real-world applications.

V. CONCLUSION

In this paper, we proposed E2Map, a spatial map that captures the agent's emotional responses to its experiences, inspired by human emotional mechanisms. By updating E2Map using various LLM capabilities, the agent can adjust its behavior in a one-shot manner after encountering specific events. Experiments in both simulated and real-world environments demonstrated that our method significantly improves navigation performance in stochastic scenarios. However, there are several promising directions for future work. Our method relies on simulator or sensor information to trigger E2Map updates. Integrating anomaly detection algorithms to autonomously identify such events would be a valuable enhancement. Furthermore, although we demonstrated that our method can also address positive emotions, our experiments primarily involved scenarios with negative emotions. Future work could explore the extension of the methodology to tasks involving positive emotions to provide a more comprehensive evaluation of its applicability. We believe that our work contributes to advancing the development of strong autonomy in robotics.

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REFERENCES

- [1] W. Huang, P. Abbeel, D. Pathak, and I. Mordatch, "Language models as zero-shot planners: Extracting actionable knowledge for embodied agents," in *9th Annual conference on machine learning*. PMLR, 2022, pp. 9118–9147.
- [2] B. Ichter, A. Brohan, Y. Chebotar, C. Finn, K. Hausman, A. Herzog, D. Ho, *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," in *6th Annual Conference on Robot Learning*, 2022.
- [3] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *6th Annual Conference on Robot Learning*, 2022.
- [4] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 9493–9500.
- [5] Y. Mu, J. Chen, Q. Zhang, S. Chen, Q. Yu, C. GE, *et al.*, "Robocodex: Multimodal code generation for robotic behavior synthesis," in *Proceedings of the 41st International Conference on Machine Learning*, ser. ICML'24. JMLR.org, 2024.
- [6] J. Chen, Y. Mu, Q. Yu, T. Wei, S. Wu, Z. Yuan, *et al.*, "Roboscript: Code generation for free-form manipulation tasks across real and simulation," *CoRR*, vol. abs/2402.14623, 2024.
- [7] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," in *7th Annual Conference on Robot Learning*, 2023.
- [8] B. Chen, F. Xia, B. Ichter, K. Rao, K. Gopalakrishnan, M. S. Ryoo, A. Stone, and D. Kappler, "Open-vocabulary queryable scene representations for real world planning," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 11 509–11 522.
- [9] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 10 608–10 615.
- [10] A. R. Damasio, *Feeling & Knowing: Making Minds Conscious*, H. Damasio, Ed. New York: Pantheon Books, 2021.
- [11] J. Tierney and R. F. Baumeister, *The power of bad: And how to overcome it*. Penguin UK, 2019.
- [12] A. Galvez-Pol, M. Nadal, and J. M. Kilner, "Emotional representations of space vary as a function of peoples' affect and interoceptive sensibility," *Scientific reports*, vol. 11, no. 1, p. 16150, 2021.
- [13] A. Zeng, M. Attarian, B. Ichter, K. M. Choromanski, A. Wong, S. Welker, *et al.*, "Socratic models: Composing zero-shot multimodal reasoning with language," in *The Eleventh International Conference on Learning Representations*, 2023.
- [14] S. S. Raman, V. Cohen, E. Rosen, I. Idrees, D. Paulius, and S. Tellex, "Planning with large language models via corrective re-prompting," in *NeurIPS 2022 Foundation Models for Decision Making Workshop*, 2022.
- [15] C. H. Song, J. Wu, C. Washington, B. M. Sadler, W.-L. Chao, and Y. Su, "Llm-planner: Few-shot grounded planning for embodied agents with large language models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2998–3009.
- [16] Y. Ding, X. Zhang, C. Paxton, and S. Zhang, "Task and motion planning with large language models for object rearrangement," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 2086–2092.
- [17] W. Yu, N. Gileadi, C. Fu, S. Kirmani, K.-H. Lee, M. G. Arenas, *et al.*, "Language to rewards for robotic skill synthesis," in *7th Annual Conference on Robot Learning*, 2023.
- [18] S. Y. Gadre, M. Wortsman, G. Ilharco, L. Schmidt, and S. Song, "Cows on pasture: Baselines and benchmarks for language-driven zero-shot object navigation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 23 171–23 181.
- [19] P. Anderson, Q. Wu, D. Teney, J. Bruce, M. Johnson, N. Sünderhauf, I. Reid, S. Gould, and A. Van Den Hengel, "Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 3674–3683.
- [20] P.-L. Guhur, M. Tapaswi, S. Chen, I. Laptev, and C. Schmid, "Airbert: In-domain pretraining for vision-and-language navigation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1634–1643.
- [21] J. Krantz, E. Wijmans, A. Majumdar, D. Batra, and S. Lee, "Beyond the nav-graph: Vision-and-language navigation in continuous environments," in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVIII 16*. Springer, 2020, pp. 104–120.
- [22] Y. Hong, Z. Wang, Q. Wu, and S. Gould, "Bridging the gap between learning in discrete and continuous environments for vision-and-language navigation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 15 439–15 449.
- [23] D. Shah, B. Osiński, S. Levine, *et al.*, "Lm-nav: Robotic navigation with large pre-trained models of language, vision, and action," in *Conference on robot learning*. PMLR, 2023, pp. 492–504.
- [24] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.
- [25] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*. MIT press, 2018.
- [26] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 2011, pp. 627–635.
- [27] H.-S. Yoon, C. Kim, S.-W. Kim, and S.-W. Seo, "Self-balancing online dataset for incremental driving intelligence," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021, pp. 4941–4946.
- [28] B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl, "Language-driven semantic segmentation," in *International Conference on Learning Representations*, 2022.
- [29] OpenAI, . A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford, *et al.*, "Gpt-4o system card," 2024.
- [30] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, *et al.*, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.
- [31] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," in *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Curran Associates, Inc., 2022, pp. 22 199–22 213.
- [32] A. M. Leventhal, "Sadness, depression, and avoidance behavior," *Behavior modification*, vol. 32, no. 6, pp. 759–779, 2008.
- [33] M. Ghorbani, Y. Liao, S. Çayköylü, and M. Chand, "Guilt, shame, and reparative behavior: The effect of psychological proximity," *Journal of Business Ethics*, vol. 114, no. 2, pp. 311–323, 2013.
- [34] M. Pivetti, M. Camodeca, and M. Rapino, "Shame, guilt, and anger: Their cognitive, physiological, and behavioral correlates," *Current Psychology*, vol. 35, pp. 690–699, 2016.
- [35] S. Dehaene, "The neural basis of the weber–fechner law: a logarithmic mental number line," *Trends in cognitive sciences*, vol. 7, no. 4, pp. 145–147, 2003.
- [36] A. Stentz, *The D* algorithm for real-time planning of optimal traverses*. Citeseer, 1994.
- [37] G. Williams, A. Aldrich, and E. Theodorou, "Model predictive path integral control using covariance variable importance sampling," *arXiv preprint arXiv:1509.01149*, 2015.
- [38] K. Koide, J. Miura, and E. Menegatti, "A portable three-dimensional lidar-based system for long-term and wide-area people behavior measurement," *International Journal of Advanced Robotic Systems*, vol. 16, no. 2, p. 1729881419841532, 2019.
- [39] Y. K. Hwang, N. Ahuja, *et al.*, "A potential field approach to path planning," *IEEE transactions on robotics and automation*, vol. 8, no. 1, pp. 23–32, 1992.
- [40] N. Koenig and A. Howard, "Design and use paradigms for gazebo, an open-source multi-robot simulator," in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (IEEE Cat. No.04CH37566)*, vol. 3, 2004, pp. 2149–2154 vol.3.